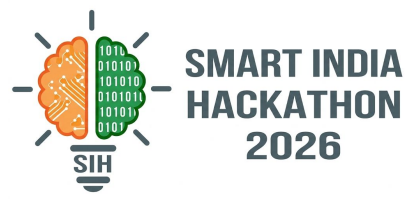
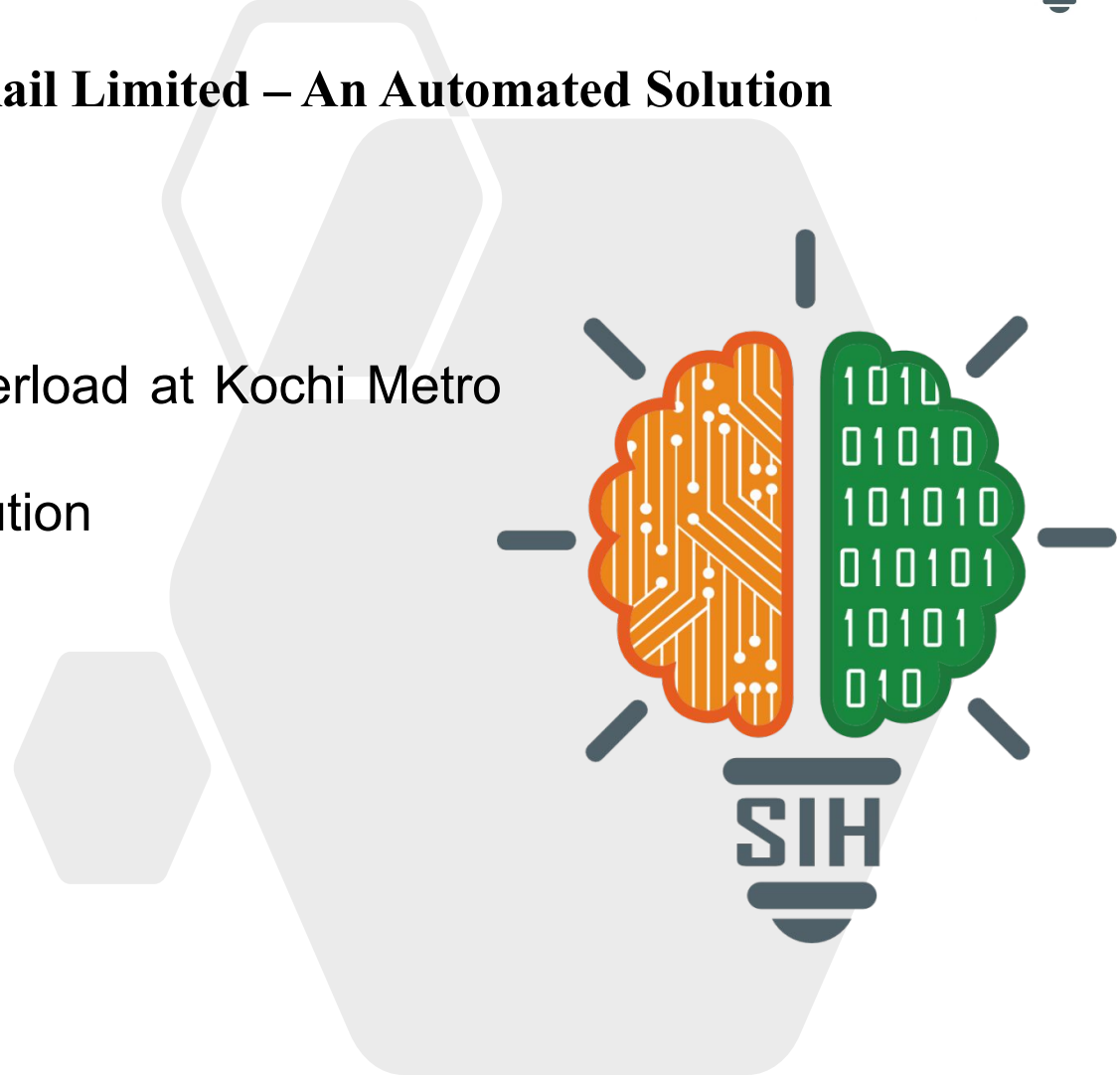


SMART INDIA HACKATHON 2026



Document Overload at Kochi Metro Rail Limited – An Automated Solution

- **Problem Statement ID** – SIH25080
- **Problem Statement Title**- Document Overload at Kochi Metro Rail Limited (KMRL) — An Automated Solution
- **Theme** - Smart Automation
- **PS Category** - Software
- **Team ID** -



PROPOSED SOLUTION :

- ❖ **AI-Powered Document Ingestion** – Automatically ingest scanned PDFs, drawings, text documents, and other KMRL records into one platform.
- ❖ **Bilingual OCR & AI Parsing** – Extract **Tamil and English text**, entities, financial amounts, authorities, deadlines, and other critical metadata.
- ❖ **AI Summarization & Risk Detection** – Generate executive briefs, actionable directives, and classify documents as **P1 Urgent, P2 High, or P3 Routine**.
- ❖ **Intelligent Department Routing** – Automatically classify and route documents to the appropriate department and track them through a **5-stage approval workflow**.
- ❖ **Semantic DeepSearch & RAG** – Search thousands of SOPs, circulars, and technical documents using natural-language questions with **clause-level source citations**.
- ❖ **SLA Monitoring & Duplicate Detection** – Track deadlines with automated escalation and detect duplicate/revised circulars using **TF-IDF + cosine similarity**.
- ❖ **Digital Seal & Audit Trail** – Complete approved documents with **SHA-256 cryptographic sealing**, role-based authorization, and tamper-resistant audit tracking.



WHY IT MATTERS ?

- ❖ **Faster & Safer Operations:** Reduces document processing delays and helps identify urgent **CMRS safety directives, deadlines, and required actions** before they are missed.
- ❖ **Less Manual Work & Better Accountability:** Automates duplicate detection, document routing, search, SLA escalation, and approval tracking while maintaining a **tamper-resistant digital audit trail**.

TECHNICAL APPROACH :

- ❖ **Document Ingestion** – Collect scanned PDFs, drawings, text documents, and other KMRL records through the document ingestion layer.
- ❖ **Bilingual AI Parsing** – Perform **Tamil/English OCR**, detect text, and extract entities such as government orders, financial amounts, authorities, and statutory deadlines.
- ❖ **AI Triage & Summarization** – Generate **3-tier summaries**, extract actionable directives, and classify documents into **P1 Urgent, P2 High, or P3 Routine**.
- ❖ **Intelligent Routing & Workflow** – Automatically assign documents to the appropriate department and move them through the **5-stage workflow**: Ingest → OCR → AI Triage → HoD Review → E-Sign & Seal.
- ❖ **Semantic Search & RAG** – Build an indexed document knowledge base and provide natural-language answers with **clause-level source citations** across technical drawings, SOPs, and circulars.
- ❖ **SLA Monitoring & Deduplication** – Monitor document deadlines with **4-tier escalation** and detect duplicate/revised circulars using **TF-IDF + Cosine Similarity**.
- ❖ **Digital Seal & Audit** – Complete approved documents with **SHA-256 HMAC digital sealing**, role-based authorization, and a tamper-resistant audit trail.

KMRL DOCUMENTS



Emails



SharePoint



Maximo



PDFs /
Scanned Files

DOCUMENT PROCESSING



OCR



Text Extraction



Metadata Extraction

KNOWLEDGE BASE



- Embeddings
- Vector Database
- RAG

AI ANALYSIS



- Summarization
- Priority Detection
- Deadline & Action Extraction

AI ASSISTANT



"What needs my attention today?"



AI → 3 urgent documents + 2 deadlines

- Text + Voice
- Question Answering
- Source References

OUTPUT

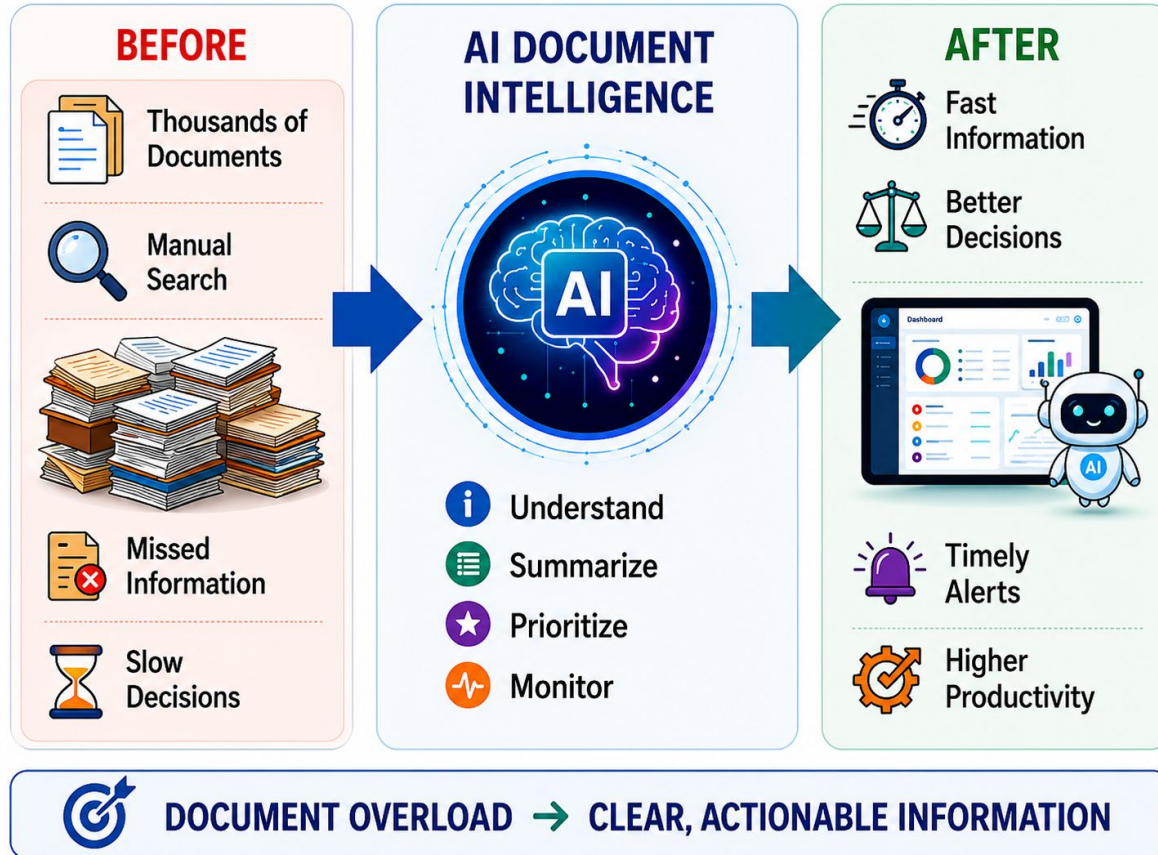


- Visual Dashboard
- Alerts, Reminders
- Actionable Insights

FEASIBILITY AND VIABILITY

- ❖ **Technical Feasibility** – Built using proven technologies including **React, Vite, Node.js, Express, OCR, NLP, TF-IDF, RAG, and SHA-256 HMAC**.
- ❖ **Modular Architecture** – Separate modules for **OCR, AI triage, summarization, workflow, RAG search, SLA monitoring, and deduplication** make the system easier to develop and maintain.
- ❖ **Bilingual Processing** – Supports **Tamil and English** document processing for Government Orders, sanctions, and administrative records.
- ❖ **AI-Assisted Automation** – Automates **classification, summarization, priority detection, entity extraction, routing, and action identification**.
- ❖ **Enterprise Search** – Inverted indexing and RAG enable fast retrieval of information from large collections of **SOPs, circulars, drawings, and technical documents**.
- ❖ **Workflow Integration** – Documents move through a structured **5-stage workflow** from ingestion to final e-sign and digital sealing.
- ❖ **Security & Compliance** – **RBAC, ACID transactions, SHA-256 HMAC sealing, backups, and audit trails** provide secure and traceable document management.
- ❖ **Tested Prototype** – The project includes **31 automated tests** covering API, database, AI/triage, deduplication, Tamil detection, and digital signatures.
- ❖ **Scalable Deployment** – The architecture can expand from the prototype to multiple KMRL departments and operational areas, including **Metro and Water Metro**.
- ❖ **Measurable Operational Value** – The prototype targets reduction of document triage from **4.2 hours to 18 seconds**, along with automated SLA tracking and duplicate detection.

IMPACT AND BENEFITS



- ❖ **Faster & Safer Operations** – Reduces document processing delays and enables rapid identification of **P1 safety directives, deadlines, and required actions**.
- ❖ **Better Decision-Making** – Provides **AI summaries, relevant documents, priorities, and actionable information** in one place.
- ❖ **Reduced Manual Work** – Automates **document classification, routing, duplicate detection, and SLA monitoring**, reducing repetitive administrative tasks.
- ❖ **Improved Compliance & Accountability** – Tracks approvals, escalations, and deadlines while maintaining a **cryptographic audit trail** through SHA-256 digital sealing.
- ❖ **Faster Information Retrieval** – Semantic RAG search enables users to find relevant information across technical drawings, SOPs, and circulars using natural-language questions.
- ❖ **Bilingual Accessibility** – Automated Tamil and English processing helps handle Government Orders and administrative documents more efficiently.

❖ **Document AI** – *LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking* — Huang et al., 2022. Useful for understanding document layout, forms, tables and visual document content.

[Research Paper – LayoutLMv3](#)

❖ **RAG** – *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks* — Lewis et al., 2020. Foundation for retrieving relevant document information before generating an answer.

[Research Paper – RAG](#)

❖ **GraphRAG** – *From Local to Global: A Graph RAG Approach to Query-Focused Summarization* — Edge et al., 2024. Supports connecting information across large collections of documents.

[Research Paper – GraphRAG](#)

❖ **Voice AI** – *Robust Speech Recognition via Large-Scale Weak Supervision* — Radford et al., 2022. The research behind Whisper-style robust speech recognition for voice interaction.